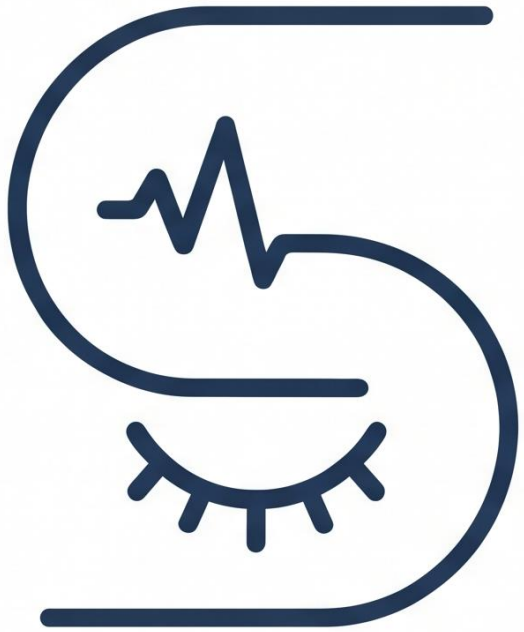
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# SWEM

FOCUS STRESS SLEEP

## Laboratory of Nearable Technologies for Health Data Science

2026 Final Project


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Beatrice Nogara

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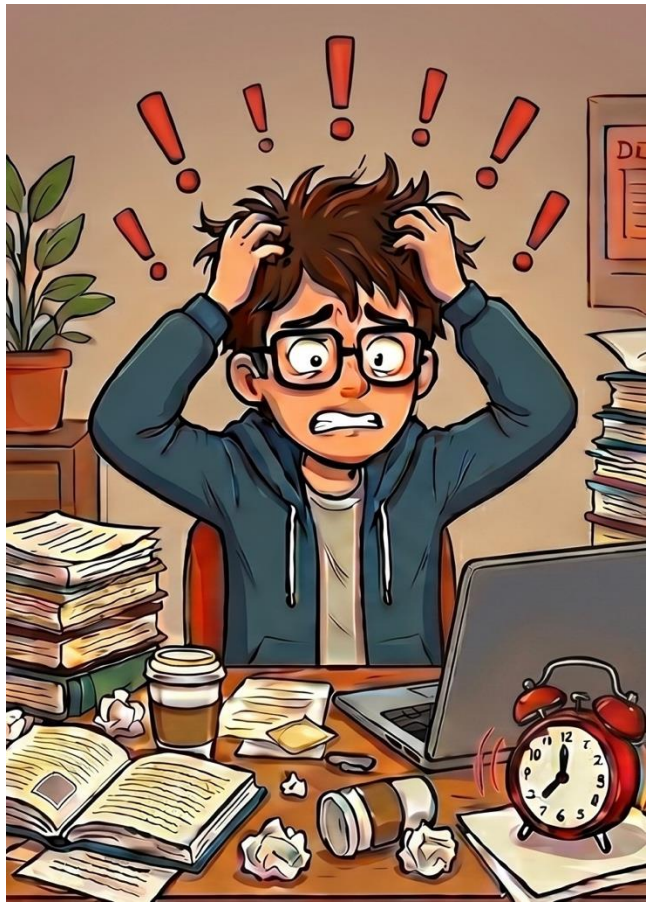
Alessia Radicula

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# Have you ever felt Stressed or Sleepy while studying?





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# Student Wellbeing Evaluation & Monitoring

## We are able to:

Detect if the student behavior is:
- Focused (normal)
- Unfocused (anomalous)

Address the root cause of anomalous behavior:
- Stressed
- Sleepy

Thanks to this analysis, the student can track their **mental wellbeing** over time and build healthier, data-driven daily routines.

## We are using:

Nearable device: **Laptop webcam**
- Posture analysis
- Blinking analysis

Wearable device: **PPG** to track the Heartbeat

**ML models** to analyze the features and formulate the output:
- Supervised (binary and 3 classes)
- Unsupervised (binary and 3 classes)

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# Posture Analysis using SWEM

The **goal** is to detect ergonomic anomalies linked to stress or sleepiness

We monitor the posture using only the laptop's **frontal webcam**

Parameters:

- Grounded in clinical literature
- Computed from MediaPipe's 33 body landmarks



**OR**



| Parameter | What it measures | Threshold | Landmark |
|---|---|---|---|
| **Shoulder Asymmetry** | Tilt of shoulder line vs horizontal | > 7° = bad | 11, 12 (left shoulder, right shoulder) |
| **Head Lateral Tilt** | Head sideways inclination vs shoulders | > 10° = bad | 7, 8, 11, 12 (left ear, right ear, left shoulder, right shoulder) |
| **Forward Head Proxy** | Nose-shoulder vertical ratio | < 0.7 = slouching | 0, 11, 12 (nose, left shoulder, right shoulder) |

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# Posture Analysis using SWEM

For each 1-minute window → 9 features extraction

**Mean**
Average value over the window

**Std**
Variability over the window

**Bad_pct**
% of frames over the threshold

| Situation | Posture Quality | Mean | Std | Bad_pct |
|---|---|---|---|---|
| Focus | Good | Low angles / high FHP ratio | Low | Low |
| Sleepy | Bad - Static | High angles / low FHP ratio | High | Low |
| Stressed | Bad - Dynamic | High angles / low FHP ratio | High | High |

*The Label is not assigned based on posture alone: this result is merged with blinking and heart rate features to train a single classification model on the combined data.*

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# Blinking Analysis using SWEM

The **goal** is to detect ergonomic anomalies linked to stress or sleepiness

We monitor the blinking using only the laptop's **frontal webcam**

Parameters evaluated with 12 facial landmarks



OR

| Parameter | What it measures | Threshold |
|---|---|---|
| **EAR** | Average eye opening | 0.25 – 0.35 = eye open |
| **PERCLOS** | Percentage of time eyes are closed more than 80% | < 0.15 = normal |
| **Blink rhythm** | Number of blink/min | 10 – 20 blink/min = normal |
| **IBI** | Time between two consecutive blink | 3.5 s = normal |

# Blinking Analysis suing SWEM

For each 1-minute window → 7 features extraction

## Focus



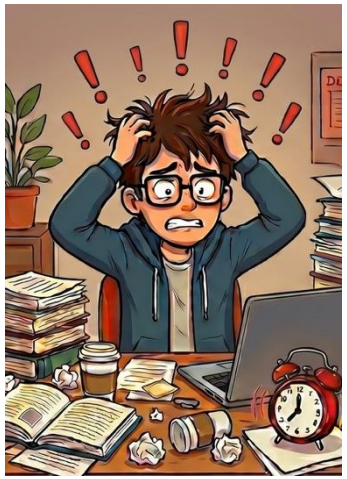
- Normal eye opening
- Regular blink rhythm
- Stable IBI pattern

## Sleepy



- Lower EAR or longer closures
- Higher PERCLOS
- Slower or prolonged blinks

## Stressed



- More variability
- Irregular blink timing
- Higher blink frequency

*The Label is not assigned based on blinking alone: this result is merged with posture and heart rate features to train a single classification model on the combined data.*

# Heart Rate Analysis using SWEM

The **goal** is to detect ergonomic anomalies linked to stress or sleepiness

We monitor the Heart Rate parameters by using a **photoplethysmogram (PPG) sensor** as a wearable device.

Parameters:

- The HR was obtained directly from the sensor
- HRV was calculated from the time intervals between consecutive pulses.



OR



| Parameter | What it measures | Focused Threshold | Acquisition |
|---|---|---|---|
| **Heart Rate** | Number of heart beats per minute. | 60–85 bpm | Extracted directly from the sensor |
| **HRV (RMSSD)** | Short-term beat-to-beat variability between consecutive heartbeats. | 25–50 ms | Calculated from the waveform |

# Heart Rate Analysis using SWEM

For each 1-minute window → 2 features extraction

**HR**
Average HR value over the window

**RMSSD**
Short-term variability of RR intervals over a time window

| Situation | HR | RMSSD | Remarks |
|---|---|---|---|
| Focus | Middle | Middle | Values related to the defined threshold |
| Sleepy | Low | High | Sensible to a strong relaxation state |
| Stressed | High | Low | Sensible to physical activity |

*The Label is not assigned based on heart rate alone: this result is merged with blinking and posture features to train a single classification model on the combined data.*

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# Data acquisition – Blinking and posture

**12 videos**
4 team members x 3 categories

**4 Sleepy**



**4 Focused**



**4 Stressed**



# Data acquisition – Heart Rate

**12 Measurements**
4 team members x 3 categories
Results in csv files




**4 Sleepy**

- 1 Measurement prior going to sleep
- 3 Measurements augmented from focused

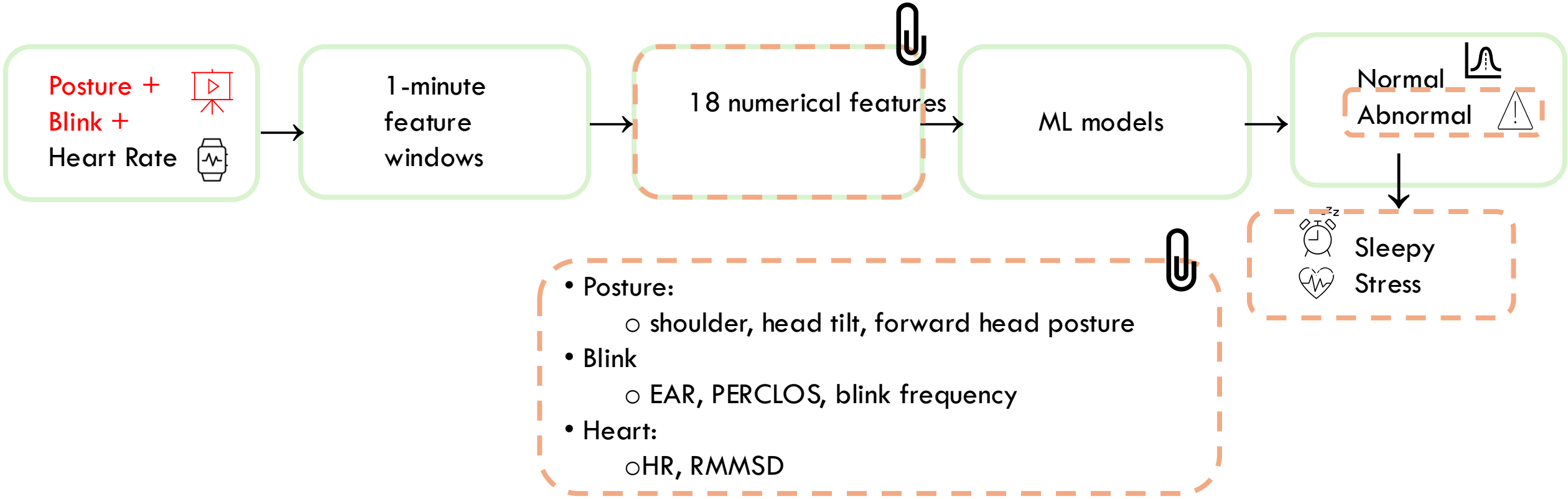
**4 Focused**

4 Measurements on normal conditions

**4 Stressed**

- 1 Measurement post exercise + energetic drink
- 3 Measurements augmented from focused

# ML Pipeline


Posture +
Blink +
Heart Rate
1-minute feature windows
18 numerical features
ML models
Normal
Abnormal
Sleepy
Stress
• Posture:
○ shoulder, head tilt, forward head posture
• Blink
○ EAR, PERCLOS, blink frequency
• Heart:
○HR, RMMSD

# ML models and evolutions

| Supervised | Unsupervised |
|---|---|
| Logistic Regression | K-means |
| SVM | Gradient Boosting |
| Random Forest | Agglomerative |
| Gradient Boosting | Isolation Forest / LOF / One-Class SVM |

**Leave-One-Subject-Out Cross-Validation**

- The dataset contains data from **4 participants.**
- In each fold, the model was trained on **3 participants.**
- The remaining **1 participant was used only for testing.**
- This evaluates whether the model can generalize to an unseen person.

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# Supervised Results

| Task | Model / Combination | Accuracy | Balanced Acc. | Macro F1 |
|---|---|---|---|---|
| Binary: Normal vs Abnormal | Logistic Regression | 0.888 | 0.915 | 0.882 |
| Binary: Normal vs Abnormal | SVM | 0.926 | 0.942 | 0.921 |
| Binary: Normal vs Abnormal | Random Forest | 0.957 | 0.958 | 0.952 |
| Binary: Normal vs Abnormal | Gradient Boosting | 0.948 | 0.948 | 0.942 |
| Direct 3-Class | Logistic Regression | 0.929 | 0.929 | 0.93 |
| Direct 3-Class | SVM | 0.895 | 0.896 | 0.892 |
| Direct 3-Class | Random Forest | 0.951 | 0.951 | 0.951 |
| Direct 3-Class | Gradient Boosting | 0.921 | 0.922 | 0.922 |
| Subtype: Sleepy vs Stress | Logistic Regression | 0.972 | 0.972 | 0.972 |
| Subtype: Sleepy vs Stress | SVM | 0.923 | 0.923 | 0.923 |
| Subtype: Sleepy vs Stress | Random Forest | 0.99 | 0.99 | 0.99 |
| Subtype: Sleepy vs Stress | Gradient Boosting | 0.986 | 0.986 | 0.986 |
| Hierarchical | RF -> RF | 0.957 | 0.957 | 0.957 |
| Hierarchical | RF -> Logistic Regression | 0.949 | 0.949 | 0.95 |
| Hierarchical | RF -> Gradient Boosting | 0.949 | 0.949 | 0.949 |
| Hierarchical | Gradient Boosting -> RF | 0.948 | 0.948 | 0.948 |
| Hierarchical | Gradient Boosting -> Gradient Boosting | 0.94 | 0.94 | 0.94 |
| Hierarchical | SVM -> RF | 0..926 | 0.926 | 0.927 |
| Hierarchical | Logistic Regression -> RF | 0.888 | 0.887 | 0.89 |

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# Unsupervised Results

| Experiment | Model | Accuracy | Balanced Acc. | Macro F1 |
|---|---|---|---|---|
| Binary Clustering | K-Means | 0.783 | 0.836 | 0.78 |
| Binary Clustering | Gaussian Mixture | 0.814 | 0.86 | 0.81 |
| Binary Clustering | Agglomerative | 0.775 | 0.831 | 0.773 |
| 3-Class Clustering | K-Means | 0.778 | 0.778 | 0.781 |
| 3-Class Clustering | Gaussian Mixture | 0.821 | 0.822 | 0.815 |
| 3-Class Clustering | Agglomerative | 0.775 | 0.775 | 0.779 |
| Anomaly Detection | Isolation Forest | 0.983 | 0.975 | 0.981 |
| Anomaly Detection | One-Class SVM | 0.974 | 0.961 | 0.97 |
| Anomaly Detection | Local Outlier Factor | 0.986 | 0.979 | 0.984 |

# Data Augmentation Experiments

Method: Gaussian Jitter

$x_aug^{(k)} = clip(x + epsilon_k)$

$epsilon_k \sim N(0, noise_level \times feature_std) \quad k = 1, ..., augmentation_factor$

| Expriment | Baseline F1 | Best Aug. F1 | Change | Best Acc. | Best Bal. Acc. |
|---|---|---|---|---|---|
| Supervised Binary classification | 0.952 | 0.957 | +0.005 | 0.961 | 0.963 |
| Supervised Hierarchical classification | 0.957 | 0.960 | +0.003 | 0.960 | 0.960 |
| Supervised Direct 3-class classification | 0.951 | 0.957 | +0.006 | 0.957 | 0.957 |
| Unsupervised Binary clustering, Gaussian Mixture | 0.810 | 0.871 | +0.061 | — | 0.907 |
| Unsupervised 3-class clustering, Gaussian Mixture | 0.815 | 0.850 | +0.035 | — | 0.851 |
| Unsupervised LOSO anomaly detection, Isolation Forest | 0.806 | 0.765 | -0.041 | — | 0.739 |

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# Sensor Ablation: do we need all the 3 sensors?

The **goal** is to quantify how much performance we lose if one or two sensors are missing

Setup:

- 3 sensors → 18 features: Posture (9), Blink (7), Heart (2)
- Re-run the pipelines keeping only sensor subsets: 3 pairs
- Models: **Random Forest** (sup.) & **Gaussian Mixture** (unsup.)
- Validation: Leave-One-Subject-Out (LOSO)
- Metric: macro-F1 (equal weight per class → robust to imbalance)



| Scenario | What happens |
|---|---|
| Full | Train + test on all 18 features (baseline) |
| Masked | Train on 18, but median-impute the missing ones at test |

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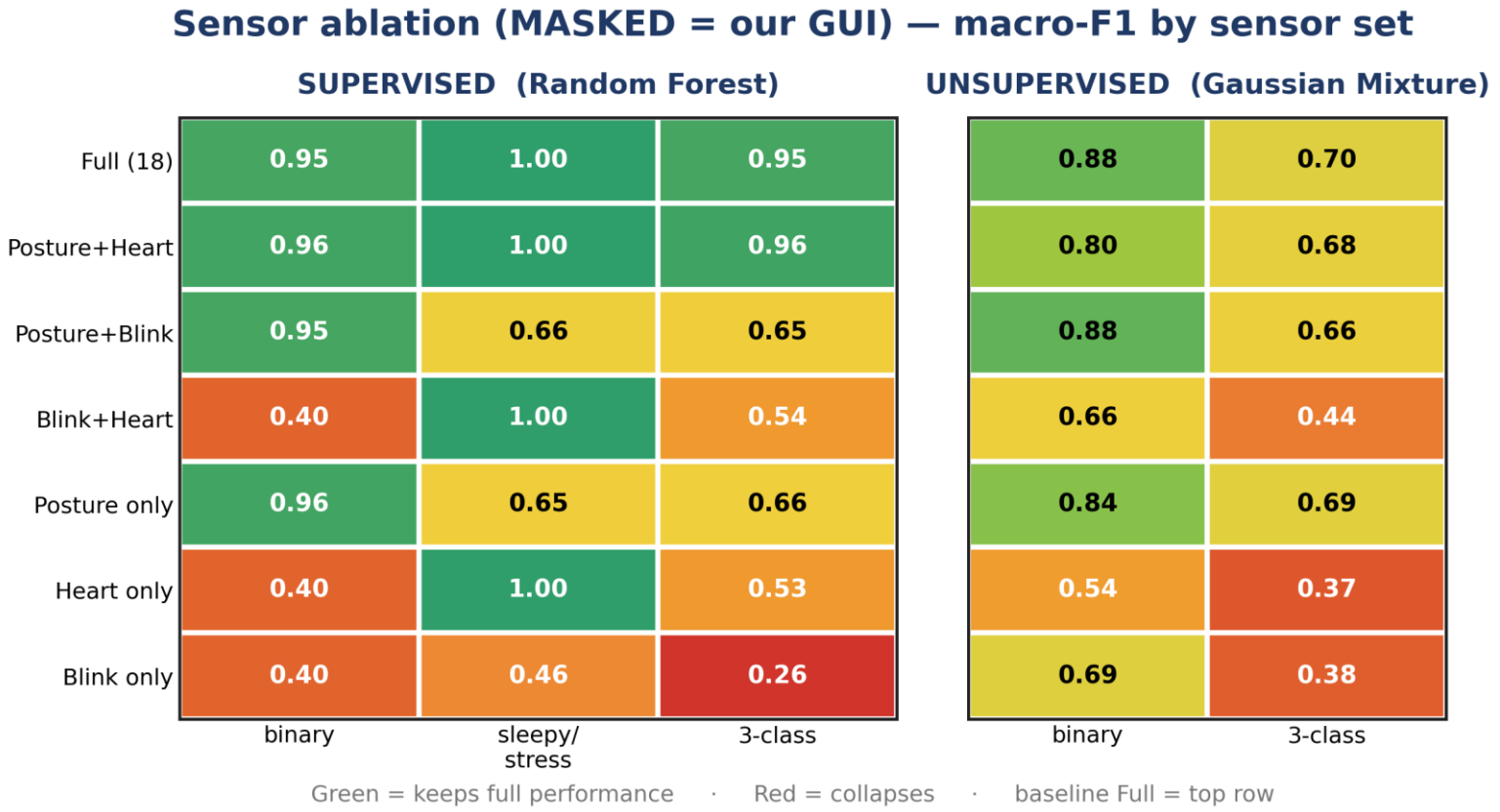
# Sensor Ablation: Takeaways

**Posture** ⟶ detects *that* something is wrong
**Heart** ⟶ detects what (sleepy vs stress)
**Blink** ⟶ droppable

**Posture + Heart ≈ full system**
**Masked Scenario** ⟶ can collapse

Supervised and Unsupervised Agree

**Deployment note (masked = our GUI)**
Safe to lose Blink or Heart (Δ ≈ 0).
**Posture is mandatory**
Fallback: a model retrained without Posture.



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# GUI Options

The user can **configure** the analysis before running the model



## Input mode

- Upload a video
- Real-time acquisition



## Classification settings

- 2 classes
- 3 classes



## Model choice

- Supervised model
- Unsupervised model



## Best model option

- Automatic model selection
- based on confusion-matrix performance

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# GUI Options

The system **combines** the selected settings and return the predicted contition

**Input Source**
Video/Real-time



**Class Setup**
2 / 3 classes



**Model Selection**



**Run Prediction**



Display Results



**Output condition**

**Focused**

**Sleepy**

**Stressed**

**2-class mode:** Normal / Anomalous

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From now on, you will remain focused!

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# Literature Review

**Posture:**

- *Scoliometer: An Overview* — ScienceDirect
- *Postural Asymmetry Analysis via Deep Learning for Scoliosis Assessment* — Springer (2024)
- *Ergonomic Assessment Norms (CUELA System)* — BMJ Open (2016)
- *Real-Time Posture Monitoring for Sitting with MediaPipe* — Springer (2025)

**Blinking:**

- *Operational Definitions of Driving Performance Measures and Statistics – SAE international Mobility Advanced (2023)*
- *Real-Time Eye Blink Detection using Facial Landmarks - Tereza Soukupova and Jan Cech (2016)*
- *PERCLOS: A Valid Psychophysiological Measure of Alertness As Assessed by Psychomotor Vigilance - David F. Dinges (1998)*
- *Blinking – Cleveland Clinic (2023)*

**Heart Rate:**

- *An Overview of Heart Rate variability Metrics and Norms — Shaffer (2017)*
- *Thermal Comfort and Stress Recognition in Office Environment — Nkurikiyeyezu (2019)*
- *Stress and Heart Rate Variability: A Meta-Analysis and Review of the Literature — Hye-Geum (2018)*
- *Reference Ranges for Short-Term Heart Rate Variability Measures in Individuals Free of Cardiovascular Disease: The Multi-Ethnic Study of Atherosclerosis (MESA) — Wesley (2016)*

# Literature Review

**ML:**


- *Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? NeurIPS Datasets and Benchmarks.*
- *Shwartz-Ziv, R., & Armon, A. (2021). Tabular Data: Deep Learning is Not All You Need. arXiv.*
- *Ruff, L., et al. (2020). A Unifying Review of Deep and Shallow Anomaly Detection. arXiv.*
- *Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.*
- *Breiman, L. (2001). Random Forests. Machine Learning.*
- *Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.*

# Annex



# Sensor Ablation: Clean VS Masked

